

Milestone 3: Full Dashboard Development (Weeks 5–6)

Objective

Milestone 3 focused on developing and integrating four fully interactive dashboards using Power BI to analyze global military strength across 140+ countries.

This phase transformed the engineered dataset (`military_final.csv`) into a dynamic analytical system with filters, navigation, and cross-dashboard interaction.

Module 5: Quick Stats & Nation Overview Development

Quick Stats Dashboard

Purpose

Provide a high-level global summary of military strength indicators.

Key Features Implemented

- KPI Cards:
 - Total Countries Analyzed
 - Global Population
 - Total Active Military Personnel
 - Total Military Assets
- Top 10 Countries by Power Index Rank Gap
- NATO vs Non-NATO Distribution (Pie Chart)
- Defense Budget-to-GDP Ratio (World Map)
- Top Countries by Total Military Assets (Bar Chart)

Interactions

- Filters:
 - Region
 - Continent
 - Alliance (NATO / Non-NATO)
- Dynamic KPI updates based on slicer selection
- Cross-filtering enabled across visuals

Outcome

Users can:

- Quickly identify global leaders
 - Analyze regional military dominance
 - Understand alliance distribution patterns
-

Nation Overview Dashboard

Purpose

Provide a detailed military profile of an individual country.

Key Features Implemented

- KPI Cards:
 - Total Manpower Strength
 - Total Air Power Assets
 - Land Force Strength
 - Overall Military Strength Index
- Visualizations:
 - Manpower Strength by Country
 - Air Power Distribution
 - Naval Power Strength
 - Defense Budget Comparison
 - Global Land Power Map
- Single Country Slicer for dynamic selection

Functionality

- Selecting a country updates:
 - All KPI cards
 - Charts and comparisons
- Tooltips provide deeper metric insights

Outcome

Users can:

- Analyze a country's military structure
 - Understand strengths across land, air, and naval divisions
 - Evaluate budget allocation and strategic positioning
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Module 6: Compare Powers & Coalition Builder Development

Compare Powers Dashboard

Purpose

Enable side-by-side military comparison between two countries.

Key Features Implemented

- Dual Country Selectors (Country1 & Country2 tables)
- Comparative Visualizations:
 - Manpower Comparison
 - Defense Budget Comparison
 - Attack Aircraft Strength
 - Population Comparison
 - Power Rank Comparison

Measures Created

Example:

```
Selected Manpower Country1 =  
CALCULATE(  
    SUM(military_final[active_personnel]),  
    military_final[country] = SELECTEDVALUE(Country1[country])  
)
```

Difference measures were calculated to show strength gaps between countries.

Outcome

Users can:

- Directly compare military capabilities
- Identify strength differences
- Analyze strategic advantage across metrics

Coalition Builder Dashboard

Purpose

Simulate military alliances and compare coalition strength against a reference country.

Key Features Implemented

- Multi-country coalition selector
- Reference country selector
- Aggregated coalition totals
- Comparison visuals:
 - Aircraft Strength Analysis
 - Defense Budget Comparison
 - Manpower Analysis
 - Naval Fleet Strength Analysis

Key Measures

Coalition Measure Example:

```
Coalition Manpower =  
SUM(military_final[active_personnel])
```

Reference Measure Example:

```
Reference Manpower =  
CALCULATE(  
    SUM(military_final[active_personnel]),  
    military_final[country] =  
        SELECTEDVALUE(ReferenceCountry[country])  
)
```

Difference Measure:

```
Manpower Difference =  
[Coalition Manpower] - [Reference Manpower]
```

Outcome

Users can:

- Build hypothetical alliances
- Compare coalition strength vs major global powers
- Analyze strategic defense positioning